

Case Study Task – Online Food Delivery Platform Analysis

Scenario

You are working as a **Data Analyst** for an online food delivery company similar to **Foodpanda/UberEats**. The management wants a **detailed analysis of customer behavior, orders, restaurants, and delivery performance** to make data-driven decisions.

You are given a dataset with ~200 rows that include information about customers, restaurants, orders, payments, and delivery times.

Step 1 – Data Preparation & Cleaning

1. Load the dataset in pandas.
2. Handle missing values (if any).
3. Convert `Order Date` into datetime format.
4. Create new calculated columns:
 - `Delivery Delay (min) = Delivery Time - Promised Time`
 - `Order Month`
 - `Day of Week`
 - `Revenue = Quantity × Price`
5. Remove duplicate orders (if any).

Step 2 – Exploratory Data Analysis (EDA)

Perform **at least 10 analyses** with pandas + numpy + seaborn/matplotlib:

- Most popular **cuisine types** (Pizza, Burger, Desi, Chinese, etc.)
- Average **delivery delay** by city
- Top 5 restaurants by **total revenue**
- Distribution of **payment methods** (Cash, Card, Wallet)
- Average **order value** by customer age groups (<20 , 20–29 , 30–39 , 40+)
- Peak **ordering days** of the week
- Monthly revenue trend
- Average **ratings** by cuisine type
- % of orders delivered **late vs on-time**
- Correlation between **delivery time & customer rating**

Step 3 – Advanced Analysis

1. Identify the **top 10 loyal customers** (highest number of orders).
2. Find which **city has the highest average order value**.
3. Detect if **premium restaurants** (rating > 4.5) generate higher revenue compared to others.
4. Analyze the effect of **delivery delay** on customer ratings (late delivery = lower ratings?).
5. Segment customers into groups (low spenders, medium, high spenders).

Step 4: Comparison

- Compare **cash vs digital payments** in terms of late delivery trends.
- Check if **older customers (40+)** prefer different cuisines compared to younger ones.

Step 5 – Visualizations

Make at least **10 different visualizations**, covering:

- Bar charts (top restaurants, cuisines, etc.)
- Line graphs (monthly revenue, daily orders)
- Pie charts (payment methods, cuisine distribution)
- Histograms (order value distribution, age groups)
- Heatmap (correlation matrix between revenue, delay, rating, etc.)

Step 6 – Business Insights Report

Write a **professional report (PDF or PPT)** including:

- Key business insights from your analysis
- Graphs and trends explained in words
- Actionable recommendations for the company, such as:
 - How to improve delivery performance
 - Which cuisines/restaurants to promote more
 - How to increase average order value

- Customer loyalty improvement strategies